Exp 12: EXPERIMENT TO DETECT OUTLIERS IN A GIVEN DATA.

AIM: Understand the procedure to identify the outliers in a given dataset.

SAMPLE CODE:

```
# Sample calculation for lower range (lr), upper range (ur), percentile
import numpy as np

array = np.random.randint(1, 100, 16)
array

array.mean()

np.percentile(array, 25)

np.percentile(array, 50)

np.percentile(array, 75)

np.percentile(array, 100)

# Outlier detection
def outDetection(array):
   sorted(array)
```

```
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
lr, ur

import seaborn as sns
%matplotlib inline

sns.displot(array)

sns.displot(array, kind='kde')

# Filter values within lower and upper limits
new_array = array[(array > lr) & (array < ur)]
new_array

sns.displot(new_array)

# Check limits for filtered array
lr1, ur1 = outDetection(new_array)
lr1, ur1

# Final array after filtering
final_array = new_array[
```

(new_array > lr1) & (new_array < ur1)

]

final_array

sns.displot(final_array)

OUTPUT:

```
#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16)  # randomly generate 16 numbers between  1 to 100
array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
array.mean()
```

```
50.5
```

```
np.percentile(array,25)
```

```
26.0
```

```
np.percentile(array,50)
```

```
56.0
```

```
np.percentile(array,75)
```

```
69.0
```

```
np.percentile(array,100)
```

```
90.0
```

```
#outliers detection
def outDetection(array):
  sorted(array)
  Q1,Q3=np.percentile(array,[25,75])
  IQR=Q3-Q1
  lr=Q1-(1.5*IQR)
  ur=Q3+(1.5*IQR)
  return lr,ur
```

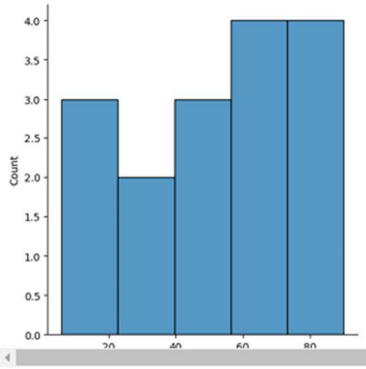
```
lr,ur=outDetection(array)
```

```
lr,ur
```

```
(-38.5, 133.5)
```

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f3291c2710>
```




```
sns.distplot(array)
```

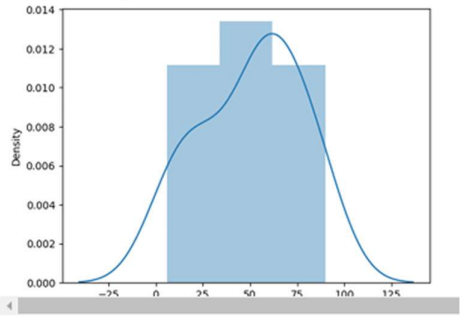
```
<ipython-input-19-d72101983c40>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(array)
<Axes: ylabel='Density'>
```

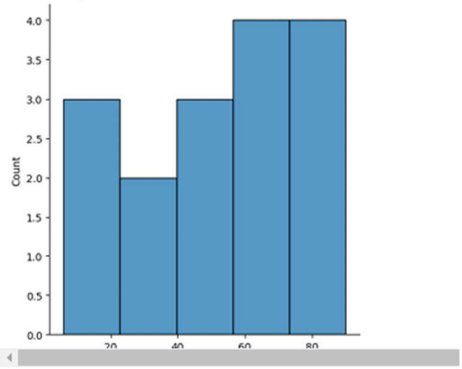



```
new_array=array[(array>lr) & (array<ur)]
new_array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
sns.displot(new_array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f2e09bb580>
```




```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

```
(-38.5, 133.5)
```

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
sns.distplot(final_array)
```

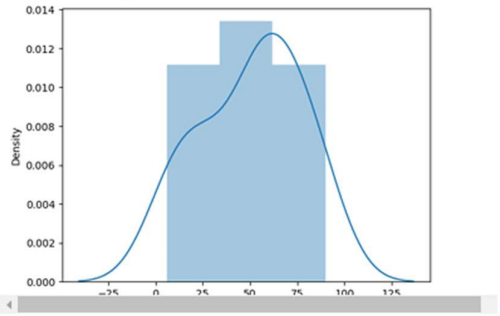
```
<ipython-input-18-7ba96ada5b76>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(final_array)
<Axes: ylabel='Density'>
```



RESULT: Thus, an experiment to detect outliers in a given data set is conducted.